# PRML PROJECT 2: TRAIN YOUR OWN GPT

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## ABSTRACT

This report follows the four components of the project specification: implementation and tuning of a GPT-style Transformer decoder with Rotary Positional Embedding (RoPE), empirical analysis of scaling and architectural modifications, parameter-efficient adaptation of a pretrained language model, and an optional study of diffusion-style language modeling. In Part A, we implement causal self-attention, RoPE, SwiGLU feed-forward layers, pre-norm residual blocks, and next-token training from core PyTorch components, then select hyperparameters by validation perplexity on Penn Treebank. The best scratch-trained configuration among the tested settings obtains a validation perplexity of $105.15_{\pm 0.93}$ at medium scale ($d_{\text{model}} = 128$, $L = 4$). In Part B, we analyze validation loss against non-embedding parameter count, token-position loss within the context window, and three architectural variants requested by the assignment: QK normalization, attention gating, and value embeddings. The parameter-scaling curve improves from the smallest to the medium model but does not support a single power-law trend across all tested sizes ($\alpha = -0.012$, $R^2 = 0.406$), while the replicated d128 ablation shows that QK normalization and attention gating reduce perplexity more reliably than value embeddings. In Part C, we fine-tune Qwen2.5-0.5B with LoRA on arXiv abstracts and compare paired generations with the base model; the adapted model eliminates examination-style formatting entirely (0/15 versus 4/15 for the base) and produces abstract-opening phrases in 8/15 continuations (versus 1/15 for the base), indicating a clear format shift, but approximately half of the adapted continuations contain garbled tokens or incoherent passages, so the format improvement is not accompanied by reliable gains in fluency or factual quality. In Part D, we compare the autoregressive GPT baseline with simplified masked-diffusion variants under matched backbone sizes, emphasizing that their losses are objective-specific and should not be interpreted as directly comparable perplexities. Across all parts, the report treats experimental claims conservatively: observed trends are separated from causal explanations, and comparisons are qualified by their training budgets, evaluation metrics, and decoding assumptions.

## 1 PART A - IMPLEMENTING A GPT WITH TRANSFORMER + ROPE

### 1.1 IMPLEMENTATION

We implement a decoder-only GPT-style language model from scratch in PyTorch and train it on the Penn Treebank (PTB) language-modeling corpus (Marcus et al., 1993). The preprocessing follows the standard word-level PTB setup: each line is whitespace-tokenized, an `<eos>` token is appended, a vocabulary is built from the training corpus, and training targets are obtained by shifting the input token sequence by one position. The model is optimized with the next-token cross-entropy objective, and evaluation perplexity is computed as $\exp(\mathcal{L})$.

The architecture is a standard pre-norm Transformer decoder: token embeddings feed into a stack of Transformer blocks, each comprising causal multi-head self-attention, a SwiGLU feed-forward network (Shazeer, 2020), and residual connections (He et al., 2016), with LayerNorm applied before each sub-layer. Positional information is encoded via Rotary Positional Embedding (RoPE) (Su et al., 2023), applied to the query and key projections prior to the attention dot product. Unlike additive absolute positional encodings, RoPE applies a position-dependent rotation to two-dimensional subspaces of the query and key representations. This makes the attention score a function of both token content and relative displacement, providing an appropriate inductive bias for causal sequence modeling without introducing a separate learned position table. A final LayerNorm and linear projection map the hidden states to vocabulary logits.

For the best-performing configuration (Experiment B), the model uses embedding dimension $d_{\text{model}} = 128$, $L = 4$ Transformer layers, $H = 4$ attention heads, head dimension $d_{\text{head}} = 32$, dropout $p = 0.1$, and maximum sequence length 256, yielding approximately 3.61M total parameters.

### 1.2 Training Setup and Hyperparameter Search

All Part A models are trained from scratch with AdamW (Loshchilov & Hutter, 2019) (weight decay $\lambda = 0.01$), gradient clipping at 1.0 (Pascanu et al., 2013), and a cosine learning-rate schedule (Loshchilov & Hutter, 2017). Model selection is based on best validation perplexity across all epochs. We evaluate five configurations at seeds 1234, 42, and 2024 (Table 1), jointly varying embedding dimension, depth, head count, learning rate, batch size, and dropout. This is a configuration sweep rather than a one-factor-at-a-time ablation.

Table 1: Part A hyperparameter sweep on PTB (10 epochs, AdamW with a cosine schedule). The five configurations jointly vary embedding dimension, depth, head count, learning rate, batch size, and dropout, and are therefore not one-factor-at-a-time ablations. Best Valid PPL is the mean ± standard deviation of the best per-epoch validation perplexity over seeds 1234, 42, and 2024; the best result (Experiment B) is shown in bold.

| Exp. | $d_{\text{model}}$ | Layers | Heads | LR | Batch | Dropout | Best Valid PPL |
|---|---|---|---|---|---|---|---|
| A | 64 | 2 | 2 | $1 \times 10^{-3}$ | 16 | 0.1 | $125.78_{\pm 0.53}$ |
| B | 128 | 4 | 4 | $1 \times 10^{-3}$ | 16 | 0.1 | $\mathbf{105.15}_{\pm 0.93}$ |
| C | 256 | 4 | 4 | $3 \times 10^{-4}$ | 16 | 0.1 | $109.28_{\pm 0.73}$ |
| D | 128 | 4 | 4 | $3 \times 10^{-4}$ | 32 | 0.2 | $178.72_{\pm 1.83}$ |
| E | 256 | 6 | 8 | $1 \times 10^{-4}$ | 16 | 0.2 | $169.97_{\pm 0.89}$ |

### 1.3 Results and Analysis

The best configuration is Experiment B ($d_{\text{model}} = 128$, $L = 4$, $H = 4$, $\eta = 10^{-3}$), with mean best validation perplexity $105.15_{\pm 0.93}$. Table 2 and Figure 1 report its seed-1234 training trajectory.

Table 2: Per-epoch training trajectory of the best Part A run (Experiment B, seed 1234). Perplexity is $\exp(\text{loss})$; the lowest validation perplexity, reached at epoch 10, is shown in bold.

| Epoch | Train Loss | Train PPL | Valid Loss | Valid PPL |
|---|---|---|---|---|
| 1 | 6.1752 | 480.66 | 5.4829 | 240.54 |
| 2 | 5.2707 | 194.55 | 5.1364 | 170.11 |
| 3 | 4.9385 | 139.56 | 4.9545 | 141.81 |
| 4 | 4.7076 | 110.79 | 4.8477 | 127.45 |
| 5 | 4.5277 | 92.55 | 4.7829 | 119.45 |
| 6 | 4.3853 | 80.26 | 4.7283 | 113.10 |
| 7 | 4.2675 | 71.35 | 4.7077 | 110.80 |
| 8 | 4.1734 | 64.94 | 4.6849 | 108.30 |
| 9 | 4.1085 | 60.86 | 4.6650 | 106.17 |
| 10 | 4.0728 | 58.72 | 4.6615 | **105.79** |

Experiment B consistently outperforms the other four tested configurations. The A–B difference is compatible with a benefit from added capacity, while the poorer C–E results may reflect some combination of architecture, learning rate, batch size, dropout, and finite training budget. Since these factors change together, the sweep cannot attribute the differences to any one of them. Validation

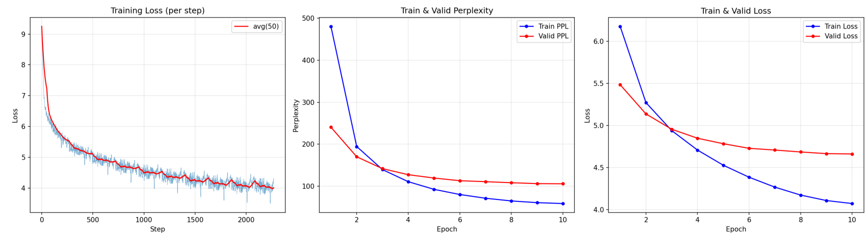



Figure 1: Loss and perplexity curves for the best Part A configuration (Experiment B, seed 1234), corresponding to the trajectory in Table 2. Training loss decreases monotonically throughout all 10 epochs; validation loss tracks closely without diverging, indicating that the model has not overfit within this training horizon.

loss for the reported seed-1234 run is still decreasing at epoch 10, so the data also do not establish that 10 epochs is optimal; a controlled extension with a fixed configuration would be required.

## 2 Part B - Scaling Laws and Architectural Study

In this part, we investigate three questions: (i) How does validation loss scale with trainable architecture parameter count under a fixed budget? (ii) How does token-position loss vary across a 256-token window? (iii) How do QK normalization, attention gating, and value embeddings behave separately and in combination? All experiments use PTB, 10 epochs, batch size 16, AdamW with cosine annealing, and gradient clipping at 1.0.

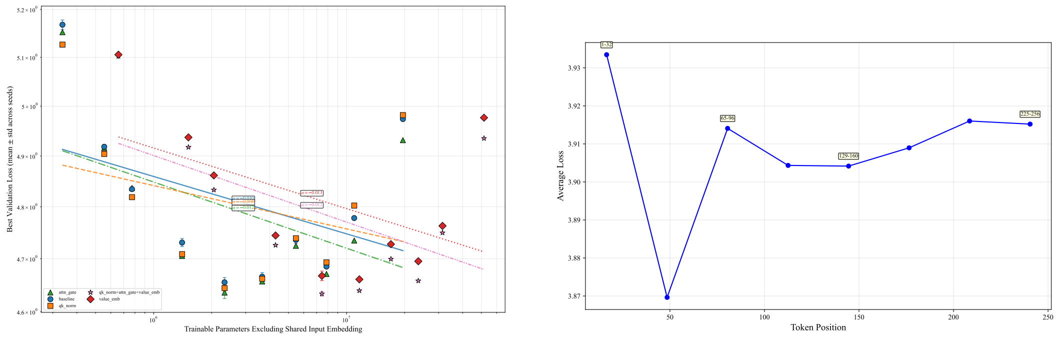
### 2.1 Scaling Laws: Loss vs. Non-Embedding Parameters

We train five baseline Transformer decoders spanning $d_{\text{model}} \in \{64, 96, 128, 160, 192\}$, depth $L \in \{2, 3, 4, 5, 6\}$, and head count $H \in \{2, 3, 4, 4, 4\}$ (Table 3), repeating each configuration at seeds 1234, 42, and 2024. We define $N_{\text{arch}}$ as all trainable parameters except the shared input token embedding. This definition includes the output projection and any architecture-specific value embeddings, while excluding only the shared input token embedding specified in the assignment. Figure 2a plots validation loss against $N_{\text{arch}}$ and fits $L(N) \propto N^{\alpha}$ in log–log space.

Table 3: Baseline configurations for the scaling study ($H$ is the attention head count). Validation loss and PPL are mean $\pm$ standard deviation over seeds 1234, 42, and 2024; $N_{\text{arch}}$ counts all trainable parameters except the shared input token embedding. The best model (s128) is shown in bold.

| Model | $d_{\text{model}}$ | $L$ | $H$ | $N_{\text{arch}}$ | Valid Loss | Valid PPL |
|---|---|---|---|---|---|---|
| s64 | 64 | 2 | 2 | 0.77 M | 4.8346$\pm$0.0042 | 125.78$\pm$0.53 |
| s96 | 96 | 3 | 3 | 1.40 M | 4.7310$\pm$0.0076 | 113.41$\pm$0.87 |
| s128 | 128 | 4 | 4 | 2.33 M | **4.6554**$\pm$0.0088 | **105.15**$\pm$0.93 |
| s160 | 160 | 5 | 4 | 3.65 M | 4.6659$\pm$0.0077 | 106.27$\pm$0.82 |
| s192 | 192 | 6 | 4 | 5.46 M | 4.7360$\pm$0.0072 | 113.98$\pm$0.82 |

The full five-point fit gives $\alpha = -0.0124$ with $R^2 = 0.406$ over 0.77–5.46M parameters, indicating that a single power law does not adequately summarize this experimental range. Restricting the fit to s64–s128 gives $\alpha = -0.0342$ and $R^2 = 0.999$ over 0.77–2.33M parameters, but this high local fit should not be extrapolated beyond the observed regime. The final two points flatten and then reverse; because model shape, optimization hyperparameters, and fixed training budget change together, we interpret the pattern as evidence of limited-budget saturation rather than as a mechanism-specific scaling law.




(a) (b)

Figure 2: Scaling and context analyses. (a) Validation loss versus trainable parameters excluding the shared input embedding; error bars show one standard deviation where three seeds are available. Lines are descriptive log–log fits, not evidence that one power law holds across the full range. (b) Cross-entropy by token-position bin for the best baseline run. Position and token composition are confounded, so the curve is observational.

### 2.2 Context Length Effect: Loss vs. Token Position

For the best baseline run (s128, seed 42, PPL $= 104.09$), we partition the 256-token context window into eight bins of 32 positions and compute mean per-token cross-entropy on the training split.

Figure 2b does not show a monotonic reduction in loss with later token position. Loss falls from 3.933 nats in positions 1–32 to 3.870 in positions 33–64, then fluctuates between 3.904 and 3.916 through the final bin. Because position is correlated with token composition and the PTB batches are contiguous slices rather than document-aligned contexts, this diagnostic should be read as an observational position profile. A controlled truncation experiment on identical target tokens would be required to isolate the marginal value of additional left context.

### 2.3 Architectural Variations

We evaluate QK normalization, attention gating, and value embeddings separately, plus an all-three combination. All variants use the same width, depth, head count, optimizer, and training budget as the corresponding baseline. The value-embedding intervention adds a separate $|\mathcal{V}| \times d_{\text{model}}$ table in every layer, so its comparison is width-matched but not parameter-matched.

**QK Normalization.** Following Dehghani et al. (2023), we apply LayerNorm to the projected query and key vectors per head before computing the usual scaled attention scores: $Q' = \text{LayerNorm}(XW_Q)$, $K' = \text{LayerNorm}(XW_K)$. This constrains query/key magnitudes and can improve optimization stability by preventing unusually large attention logits; it does not imply that standard $1/\sqrt{d_{\text{head}}}$-scaled attention necessarily sharpens with width.

**Attention Gate.** A learnable per-head scalar gate $\gamma_h$ is applied to the attention output before the output projection, following the SDPA Elementwise approach of Qiu et al. (2025): $\text{AttnOut}_h = \text{AttnOut}_h \cdot \sigma(\gamma_h)$, where $\sigma$ is the sigmoid function and $\gamma_h$ is initialized to zero. Initialization at zero sets $\sigma(\gamma_h) = 0.5$, so all heads initially contribute at half strength; the gates are then learned end-to-end to assign differential importance weights across heads, effectively providing a differentiable head-importance mechanism without hard pruning.

**Value Embedding.** Each layer has a dedicated token embedding matrix $\text{Emb}_{v,\ell} \in \mathbb{R}^{|\mathcal{V}| \times d_{\text{model}}}$ that is added to its value projection: $V_\ell = XW_{V,\ell} + \text{Emb}_{v,\ell}(\text{input\_ids})$. This adds substantial token-specific capacity: four such tables are present in the d128 model.

At matched parameter count, QK Norm reduces mean PPL by 1.15 and Attention Gate by 2.02 relative to the baseline. Value Embedding alone is 1.28 PPL worse despite adding 5.12M parameters,

Table 4: Controlled architectural ablation at the d128 baseline over three seeds; every variant shares the baseline's width, depth, head count, optimizer, and training budget. $N_{\text{arch}}$ excludes only the shared input embedding; the Value Embedding and All-three rows add per-layer value tables and are thus width-matched but not parameter-matched to the baseline. ΔPPL is the change in mean validation PPL relative to baseline (negative is an improvement); the best result is shown in bold.

| Variant | $N_{\text{arch}}$ | Valid Loss | Valid PPL | ΔPPL |
|---|---|---|---|---|
| Baseline | 2.331 M | $4.6554_{\pm 0.0088}$ | $105.15_{\pm 0.93}$ | — |
| QK Norm | 2.331 M | $4.6444_{\pm 0.0056}$ | $104.00_{\pm 0.59}$ | $-1.15$ |
| Attention Gate | 2.331 M | $4.6360_{\pm 0.0111}$ | $103.13_{\pm 1.14}$ | $-2.02$ |
| Value Embedding | 7.451 M | $4.6675_{\pm 0.0090}$ | $106.43_{\pm 0.96}$ | $+1.28$ |
| All three | 7.451 M | $\mathbf{4.6337}_{\pm 0.0045}$ | $\mathbf{102.90}_{\pm 0.46}$ | $-2.25$ |

suggesting that the added token-specific capacity is not useful under this training budget. The combined model has the best mean, but it is only 0.23 PPL better than Attention Gate while carrying the extra value tables; this difference is small relative to seed variation and should not be interpreted as evidence of a reliable interaction effect. Single-seed sweeps at the other widths are shown in Figure 2a, but the replicated d128 comparison is the primary ablation result.

# 3 Part C - Fine-Tuning a Pre-Trained Model

## 3.1 Setup

**Model.** We fine-tune Qwen2.5-0.5B (Qwen et al., 2025), a 494M-parameter decoder-only Transformer loaded in `bfloat16`. LoRA (Hu et al., 2022) with rank $r = 8$ and scaling factor $\alpha_{\text{LoRA}} = 16$ is applied to the attention and feed-forward linear projections. This introduces 4,399,104 trainable parameters ($0.89\%$); all base weights remain frozen.

**Dataset and Domain Rationale.** We use 45,000 training and 5,000 validation texts from the arXiv Abstracts dataset (Clement et al., 2019). Scientific abstracts provide a distinctive register with technical vocabulary and recurring motivation–method–result structure. Concatenating tokenized texts with an end-of-sequence separator and applying 256-token windows at 50% stride yields 59,052 training and 6,584 validation chunks.

**Training.** We use micro-batch size 4 and accumulate four micro-batches (effective batch size 16), validation batch size 16, learning rate $5 \times 10^{-5}$, 10% linear warm-up followed by linear decay, weight decay 0.01, and gradient clipping at 1.0. Three planned epochs correspond to 3,691 optimizer updates per epoch and 11,073 in total. We establish a validation baseline at step 200, skip intermediate full validations during warm-up, then evaluate every 200 updates with patience 3.

## 3.2 Results and Analysis

The validation protocol evaluates at optimizer steps 200, 1200, 1400, and 1600, obtaining validation losses 2.8797, 2.8823, 2.8852, and 2.8836. Early stopping patience is applied only after the 1,107-step warm-up, and training stops at step 1600, about $43\%$ of the first epoch. The step-200 adapter has the lowest observed validation loss and is restored before generation, so the qualitative comparison uses the selected checkpoint rather than the final training state.

We prompt both models with five identical English prefixes and draw three continuations per prefix using seed 1234 (with deterministic per-sample offsets), temperature 0.8, top-$p = 0.9$, and at most 120 new tokens. This 15-sample-per-model comparison is qualitative; it is designed to inspect register, not to estimate factual accuracy.

**Output Format and Coherence.** Table 7 summarizes the complete generation artifact rather than selected excerpts. Examination-style formatting appears in 4/15 base continuations and 0/15 adapted continuations, and the adapted model produces 8/15 continuations containing at least one pre-registered abstract marker (“in this paper,” “we present,” “we propose,” “we study,” or “our results”), compared with 1/15 for the base model. Both shifts are in the direction expected from fine-tuning on arXiv abstracts: the adapter suppresses exam-format output entirely and substantially increases the use of abstract-opening phrases. However, approximately half of the adapted continuations contain garbled tokens or non-ASCII characters (e.g., isolated Japanese characters, neologisms), indicating that the format shift is accompanied by a coherence degradation in a substantial fraction of outputs.

**Relevance to Domain.** Among the adapted continuations with coherent text, several exhibit plausible scientific register: magnetic-field analysis of star clusters, functional-analysis arguments about eigenfunctions, quantum-field theory derivations, and astrophysical spectral decomposition. These continuations use first-person plural voice and structured argument patterns consistent with arXiv abstracts. The base model also generates coherent specialized continuations—including magnetic-sensor fabrication, charged-particle measurement, and laboratory-protocol instructions—but none adopt the abstract-opening structure characteristic of research papers, and four continuations follow Chinese-curriculum exam formatting. The adapted model thus shows a qualitative shift in output register for the subset of fluent continuations, but this improvement is diluted by the garbled outputs that constitute roughly half of the adapted sample.

**Domain-Specific Vocabulary and Factuality.** The presence of abstract-marker phrases in 8/15 adapted continuations confirms a surface-level format shift, but content quality is uneven. Coherent adapted continuations deploy physics and mathematics vocabulary (e.g., gravity-wave geometry, stellar-population SEDs, non-commutative quantum mechanics), while others produce fluent but factually vague or circular prose. Several adapted outputs also contain obviously garbled passages, suggesting that the fine-tuning destabilized parts of the model's generation distribution. Since the experiment uses open-ended continuations without reference answers, factual accuracy and scientific correctness cannot be assessed from these samples alone.

Taken together, the fine-tuning experiment demonstrates a measurable format shift: the adapter eliminates exam-style output entirely and substantially increases abstract-opening phrase usage. This shift is not, however, accompanied by uniform fluency: roughly half of the adapted continuations contain token-level artifacts that undermine coherence. A stronger conclusion about domain-register acquisition would require fluency-controlled evaluation, a larger and more targeted prompt suite, blinded human judgments, and reference-based factual assessment.

# 4 Part D - Diffusion Language Model

Standard autoregressive (AR) language models generate text strictly left-to-right, each forward pass producing one token conditioned on all preceding tokens. Diffusion language models (DLMs) offer a different paradigm: masked or absorbing-state variants begin from a fully masked sequence and iteratively reveal tokens using bidirectional context. Recent work has shown that carefully designed masked diffusion objectives can provide a viable alternative to left-to-right generation while retaining flexible sampling and editing behavior (Sahoo et al., 2024; Shi et al., 2024; Nie et al., 2025). This section implements and evaluates two simplified masked DLMs on PTB, comparing them against the Part A/B GPT baseline along three axes required for a fair exploratory study: within-family loss scaling, lexical generation diagnostics, and inference dynamics.

## 4.1 Models and Experimental Setup

Three generative architectures, all sharing the same Transformer backbone (pre-norm, RoPE, SwiGLU FFN) at matched parameter counts, are compared:

- **GPT (autoregressive).** Causal self-attention; next-token cross-entropy objective; temperature-based autoregressive decoding.
- **MaskedDiffusionLM.** A simplified masked-diffusion baseline related to modern MDLM formulations (Sahoo et al., 2024; Shi et al., 2024), with bidirectional self-attention. Forward process: independently mask each token with probability $t \sim \mathcal{U}(0, 1)$. Training loss: unweighted cross-entropy on masked positions. Decoding: iterative unmasking with a linear schedule over 32 steps. This implementation is not an exact reproduction of either cited objective.

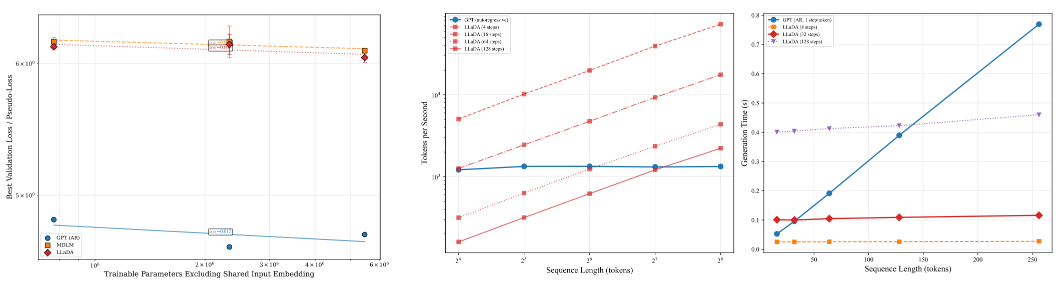
- **LLaDA** (Nie et al., 2025). Bidirectional self-attention. Forward process: mask tokens with $t \sim \mathcal{U}([0.001, 1))$. Training loss: $1/t$-weighted cross-entropy on masked positions, up-weighting difficult low-$t$ denoising steps. Decoding: cosine unmasking schedule with low-confidence remasking over 128 steps.

All three models are trained at three sizes (S: $d_{\text{model}} = 64$, $L = 2$, $H = 2$; M: $d_{\text{model}} = 128$, $L = 4$, $H = 4$; L: $d_{\text{model}} = 192$, $L = 6$, $H = 4$) for 10 epochs with AdamW and cosine annealing, at seeds 1234, 42, and 2024.

### 4.2 Scaling Behavior

Table 5: Best validation objective at three model sizes (S: $d_{\text{model}}$=64, $L$=2, $H$=2; M: $d_{\text{model}}$=128, $L$=4, $H$=4; L: $d_{\text{model}}$=192, $L$=6, $H$=4), mean $\pm$ standard deviation over seeds 1234, 42, and 2024. GPT uses next-token cross-entropy, while MDLM and LLaDA use their respective masked-prediction objectives; values are therefore comparable only within a row (across sizes), not across rows.

| Model | Valid Loss (S) | Valid Loss (M) | Valid Loss (L) |
|---|---|---|---|
| GPT (AR) | 4.835$\pm$0.004 | 4.655$\pm$0.009 | 4.736$\pm$0.007 |
| MDLM | 6.185$\pm$0.036 | 6.186$\pm$0.135 | 6.109$\pm$0.017 |
| LLaDA | 6.141$\pm$0.017 | 6.162$\pm$0.087 | 6.050$\pm$0.039 |




(a) (b)

Figure 3: (a) Validation objectives versus trainable parameters excluding the shared input embedding. Cross-family vertical offsets are not quality comparisons. (b) Measured throughput and latency on one A100-SXM4-40GB over sequence lengths 16–256. In this measured range, fixed-step LLaDA amortizes its passes over longer outputs; the result is conditional on step count and is not an asymptotic claim.

Table 5 and Figure 3a reveal two principal findings.

**Metric incomparability across paradigms.** The exponentiated DLM objectives are substantially higher than GPT perplexity, but this is not evidence that the DLMs are weaker language models. Next-token perplexity and the two masked losses evaluate different conditional distributions and use different weighting schemes. Their numerical values therefore cannot be compared across paradigms or between the two DLM objectives. We interpret only the scaling trend within each fixed model family and objective.

**Within-family trends.** GPT improves from S to M and degrades at L. The three-seed MDLM means are nearly equal at S and M and lower at L; LLaDA is also lowest at L. Standard deviations, especially MDLM-M and LLaDA-M, show that single-seed trajectories were unstable. Because masking and loss weighting differ, these results do not isolate a mechanism.

Table 6: Generation metrics at M size ($d_{\text{model}}$=128), aggregated over three seeds with five 64-token samples per seed (15 per model). TTR (type–token ratio, length-controlled) measures lexical diversity overall and per sample; bigram overlap is the mean shared-bigram fraction between sample pairs; Repetition is the fraction of samples containing a threefold $n$-gram loop. Arrows mark the preferred direction; these lexical diagnostics do not measure coherence or factuality.

| Model | TTR (overall) ↑ | TTR/sample ↑ | Bigram overlap ↓ | Repetition ↓ |
|---|---|---|---|---|
| GPT (AR) | $0.560_{\pm 0.007}$ | $0.733_{\pm 0.007}$ | $0.014_{\pm 0.004}$ | $0.000_{\pm 0.000}$ |
| MDLM | $0.514_{\pm 0.007}$ | $0.655_{\pm 0.013}$ | $0.012_{\pm 0.004}$ | $0.200_{\pm 0.000}$ |
| LLaDA | $0.534_{\pm 0.056}$ | $0.689_{\pm 0.091}$ | $0.016_{\pm 0.014}$ | $0.267_{\pm 0.231}$ |

### 4.3 Generation Quality

GPT has the highest TTR and no detected threefold $n$-gram loops in these samples. LLaDA's mean TTR is slightly above MDLM's, but it also has larger seed variation and a higher mean repetition rate. Because these diagnostics measure lexical properties rather than semantic coherence or factuality, and because the decoding procedures are not quality-matched, the table should not be read as an overall ranking of model quality.

### 4.4 Inference Dynamics

We benchmark FP32 inference on one NVIDIA A100-SXM4-40GB with batch size 4, three warm-up runs, and ten timed repetitions. GPT uses greedy decoding without a KV cache and a 256-token context window. Its measured throughput is 1,213–1,337 tok/s over lengths 16–256. CUDA is synchronized before and after every timed region.

LLaDA predicts all positions on every pass. At 128 steps, throughput rises from 160 tok/s at length 16 to 2,226 tok/s at length 256 and exceeds this uncached GPT implementation only at the longest tested length; reducing the number of diffusion steps shifts the crossover to shorter outputs. This benchmark is a systems comparison under a specific implementation, not a quality-equivalent comparison. Self-attention cost, hardware, kernel efficiency, cache usage, and the chosen diffusion-step budget all limit generalization beyond the measured range.

## Contribution Statement

Qishen Zhong was mainly responsible for Part A: implementing the GPT-style Transformer with RoPE, running the hyperparameter sweep, and producing the Part A training curves and analysis. Kaidi Zha was mainly responsible for Parts B and C: extending the architecture for the scaling-law and architectural-comparison experiments, running the Qwen2.5-0.5B LoRA fine-tuning experiment, and organizing the corresponding figures and generation examples. Both authors were responsible for Part D. Both authors discussed the experimental design of all parts, and integrated the final report.

## References

Colin B. Clement, Matthew Bierbaum, Kevin P. O'Keeffe, and Alexander A. Alemi. On the use of arxiv as a dataset, 2019.

Mostafa Dehghani, Josip Djolonga, Basil Mustafa, Piotr Padlewski, Jonathan Heek, Justin Gilmer, Andreas Steiner, Mathilde Caron, Robert Geirhos, Ibrahim Alabdulmohsin, Rodolphe Jenatton, Lucas Beyer, Michael Tschannen, Anurag Arnab, Xiao Wang, Carlos Riquelme, Matthias Minderer, Joan Puigcerver, Utku Evci, Manoj Kumar, Sjoerd van Steenkiste, Gamaleldin F. Elsayed, Aravindh Mahendran, Fisher Yu, Avital Oliver, Fantine Huot, Jasmijn Bastings, Mark Patrick Collier, Alexey Gritsenko, Vighnesh Birodkar, Cristina Vasconcelos, Yi Tay, Thomas Mensink, Alexander Kolesnikov, Filip Pavetić, Dustin Tran, Thomas Kipf, Mario Lučić, Xiaohua Zhai, Daniel Keysers, Jeremiah Harmsen, and Neil Houlsby. Scaling vision transformers to 22 billion parameters, 2023. URL https://arxiv.org/abs/2302.05442.

Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2016.

Edward J Hu, yelong shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2022. URL https://openreview.net/forum?id=nZeVKeeFYf9.

Ilya Loshchilov and Frank Hutter. SGDR: Stochastic gradient descent with warm restarts. In *International Conference on Learning Representations*, 2017. URL https://openreview.net/forum?id=Skq89Scxx.

Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations*, 2019. URL https://openreview.net/forum?id=Bkg6RiCqY7.

Mitchell P. Marcus, Beatrice Santorini, and Mary Ann Marcinkiewicz. Building a large annotated corpus of English: The Penn Treebank. *Computational Linguistics*, 19(2):313–330, 1993. URL https://aclanthology.org/J93-2004/.

Shen Nie, Fengqi Zhu, Zebin You, Xiaolu Zhang, Jingyang Ou, Jun Hu, Jun Zhou, Yankai Lin, Ji-Rong Wen, and Chongxuan LI. Large language diffusion models. In D. Belgrave, C. Zhang, H. Lin, R. Pascanu, P. Koniusz, M. Ghassemi, and N. Chen (eds.), *Advances in Neural Information Processing Systems*, volume 38, pp. 50608–50646. Curran Associates, Inc., 2025. URL https://proceedings.neurips.cc/paper_files/paper/2025/file/48b383b24230e0e6e649d9c98dae4d8c-Paper-Conference.pdf.

Razvan Pascanu, Tomas Mikolov, and Yoshua Bengio. On the difficulty of training recurrent neural networks, 2013. URL https://arxiv.org/abs/1211.5063.

Zihan Qiu, Zekun Wang, Bo Zheng, Zeyu Huang, Kaiyue Wen, Songlin Yang, Rui Men, Le Yu, Fei Huang, Suozhi Huang, Dayiheng Liu, Jingren Zhou, and Junyang Lin. Gated attention for large language models: Non-linearity, sparsity, and attention-sink-free. In D. Belgrave, C. Zhang, H. Lin, R. Pascanu, P. Koniusz, M. Ghassemi, and N. Chen (eds.), *Advances in Neural Information Processing Systems*, volume 38, pp. 100092–100118. Curran Associates, Inc., 2025. URL https://proceedings.neurips.cc/paper_files/paper/2025/file/904e89bb4e632e75fb47f093b620b257-Paper-Conference.pdf.

Qwen, :, An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chengyuan Li, Dayiheng Liu, Fei Huang, Haoran Wei, Huan Lin, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Yang, Jiaxi Yang, Jingren Zhou, Junyang Lin, Kai Dang, Keming Lu, Keqin Bao, Kexin Yang, Le Yu, Mei Li, Mingfeng Xue, Pei Zhang, Qin Zhu, Rui Men, Runji Lin, Tianhao Li, Tianyi Tang, Tingyu Xia, Xingzhang Ren, Xuancheng Ren, Yang Fan, Yang Su, Yichang Zhang, Yu Wan, Yuqiong Liu, Zeyu Cui, Zhenru Zhang, and Zihan Qiu. Qwen2.5 technical report, 2025. URL https://arxiv.org/abs/2412.15115.

Subham Sekhar Sahoo, Marianne Arriola, Yair Schiff, Aaron Gokaslan, Edgar Marroquin, Justin T. Chiu, Alexander Rush, and Volodymyr Kuleshov. Simple and effective masked diffusion language models. In *Advances in Neural Information Processing Systems*, volume 37, pp. 126354–126409. Curran Associates, Inc., 2024. URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/eb0b13cc515724ab8015bc978fdde0ad-Abstract-Conference.html.

Noam Shazeer. Glu variants improve transformer, 2020. URL https://arxiv.org/abs/2002.05202.

Jiaxin Shi, Kehang Han, Zhe Wang, Arnaud Doucet, and Michalis K. Titsias. Simplified and generalized masked diffusion for discrete data. In *Advances in Neural Information Processing Systems*, volume 37, 2024. URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/bad233b9849f019aead5e5cc60cef70f-Abstract-Conference.html.

Jianlin Su, Yu Lu, Shengfeng Pan, Ahmed Murtadha, Bo Wen, and Yunfeng Liu. Roformer: Enhanced transformer with rotary position embedding, 2023. URL `https://arxiv.org/abs/2104.09864`.

## A REPRODUCIBLE FINE-TUNING EVALUATION

Every base and adapted continuation reported in this appendix is reproduced in full; no result is reconstructed from a truncated console excerpt. The complete generation set comprises five prompts $\times$ three samples per prompt, drawn with seed 1234 (with deterministic per-sample offsets), temperature 0.8, top-$p = 0.9$, and at most 120 new tokens, for a total of 15 continuations per model.

Table 7: Format diagnostics computed from the complete 15 continuations per model (120 new tokens, temperature 0.8, top-$p = 0.9$). "Exam-format" counts continuations containing multiple-choice or answer-key markers; "Abstract-marker" counts continuations containing at least one pre-registered phrase: "in this paper," "we present," "we propose," "we study," or "our results." The adapted model eliminates exam-format output entirely and produces abstract-marker phrases in 8 of 15 continuations, compared with 1 of 15 for the base model; approximately half of the adapted continuations also contain garbled tokens. Mean words counts all words in the full generation including the prompt prefix.

| Model | Exam-format outputs | Abstract-marker outputs | Mean words |
|---|---|---|---|
| Base | 4 | 1 | 101.6 |
| LoRA adapted | 0 | 8 | 94.4 |

Table 7 is consistent with the main-text interpretation: the adapted model shows a clear format shift (zero exam-format, eight abstract-marker continuations), but this shift coexists with token-level artifacts in approximately half the adapted outputs. The following pair illustrates both the format shift and the coherence limitation. For prompt "In recent years, researchers have discovered that" (sample 0, seed 2234), the complete outputs up to the configured generation boundary are:

Base:

> In recent years, researchers have discovered that many natural medicines, including medicinal herbs, are effective in treating many diseases and ailments. One of these diseases and ailments is cancer. In this lab, you will investigate the potential for using a natural compound to treat cancer. • You will be using a computer to explore the potential of natural compounds to treat cancer. • You will be using an internet search engine to find information about the potential of natural compounds

LoRA adapted:

> In recent years, researchers have discovered that the magnetic field in the gas of a cluster of stars can be enhanced and its magnetic field variation is related to the star formation process. We propose to use the magnetic field in the gas to evaluate the star formation process in a cluster of stars. The magnetic field has been used to investigate the gravitational influence on a gas phase point of the cluster. In this paper, we use the magnetic field to analyze the growth of the gas in a cluster. We suggest that the magnetic field has been detected in the gas of the cluster. The magnetic field is considered to be an important factor in the [garbled] and gravity

The base continuation adopts a laboratory-instruction format with no abstract-style markers. The adapted continuation uses first-person plural voice and contains two pre-registered abstract markers ("We propose" and "In this paper"), consistent with arXiv abstract register; the token artifact at the end of the passage (rendered here as [garbled]) illustrates the coherence limitation noted in Section 3 for roughly half of the adapted outputs.